

Code-Layout, Readability and Reusability

Date	7 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	YES	Follows Python indentation standards
2	Proper File Structure	Files and folders are logically organized	YES	Separate templates, static files, model, and application code
3	Meaningful Variable Names	Variables reflect their purpose clearly	YES	Descriptive names such as life_expectancy, gni, predicted_hdi
4	Function / Method Names	Functions are descriptively named	YES	Functions like home() and predict() are Meaningful
5	Code Comments	Inline and block comments explain important logic	YES	Key sections are documented
6	Modular Design	Code is split into reusable modules/reusable	YES	Flask routes and model loading are separated logically
7	No Redundant Code	Duplicate or unnecessary code is removed	YES	Code is clean and optimize
8	Error Handling	Exceptions and errors are handled gracefully	PARTIAL	Basic validation implemented; can be enhanced further

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Flask Application (app.py)	Python, Flask	Handles routing and prediction requests	High

2	Machine Learning Model (HDI.pkl)	Scikit-learn	Used for HDI prediction	High
3	Input Form (index.html)	HTML, CSS	User data entry	High
4	Result Page (result.html)	HTML, CSS	Displays prediction results	High
5	Prediction Logic	Python, Pandas	Data preparation and model prediction	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate folders and modules
Readability	5	Clear naming conventions and easy-to-understand code
Reusability	5	Components such as model, templates, and prediction logic can be reused
Documentation / Comments	5	Important sections are commented; additional documentation can be added
Overall Score	5	Clean, modular, maintainable, and suitable for future enhancements